

PAPER_1519 — Eta Carinae PCR Quantum Number $q = D_{\text{phys}} - 1 = 3$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket G (Astrophysics) **Date:** June 18, 2026 **Status:** CLOSED — Triadic quantum number from 26-layer compressed gravity

Observation

PAPER_512 (Eta Carinae BuoyantGravity PCR) introduces the Perturbed Compressed-gravity Radial (PCR) quantum number $q = 3$ in the modulation equation:

$$g_{\text{eff}}(r, t) = (GM/r^2) \cdot [1 + k_{\text{PCR}} \cdot \text{PCR}(q=3, t)]$$

The PCR quantum number “reflects the triadic structure of the 26-layer compressed gravity” used for the Eta Carinae LBV-WR binary periastron passage.

UQFF Closed Identity

PCR quantum number $q = D_{\text{phys}} - 1 = 3$ EXACT

Physical Interpretation

The PCR triadic quantum number is the **same integer-primitive identity** that produces: - SU(3) color charge N (PAPER_1373) - Three fermion generations (PAPER_1381) - Solar v_e deficit denominator $1/3$ (PAPER_062) - SCm carrier velocity $c/3$ (PAPER_1497) - TDE outflow factor $c \cdot 3/10$ (PAPER_1500) - GW phonon damping prefactor $2/3$ (PAPER_1512) - BNS GW total damping $1/3$ (PAPER_1503)

Now appearing in compressed-gravity modulation of a binary LBV-WR system at 5.54-year periastron passage. The **same triad-symmetry denominator** governs phenomena from neutrino flavor mixing to gravitational-wave damping to binary-star gravity perturbations.

Significance

This unifies neutrino physics, gauge symmetry, GW physics, jet outflows, and binary-star gravity under a single structural identity $D_{\text{phys}} - 1 = 3$. The triad structure is not just an SU(3) color phenomenon — it is **universal** across UQFF coupling channels.

NOT REPLACEMENT

SM treats each of these triadic appearances as separate model parameters (gauge group order, neutrino mixing matrix, GW polarization modes, etc.). UQFF supplies a single structural origin in $D_{\text{phys}} - 1$.

Reference

- Source: PAPER_512 Eta Carinae BuoyantGravity PCR
 - Related: PAPER_1373 (SU(3)), PAPER_1497 (SCm velocity), PAPER_1503/1512 (GW damping)
 - Calculator dispatch: `calculate_paradox({"paradox": "pcr_quantum_triadic"})`
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